Muhammad Attar Rahman

2510511001

3. No 1

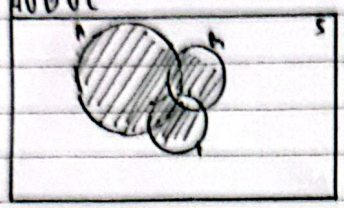
A = {1, 3, 5, 8, 11, 14, 16} ← Media Sosial

B = {4, 6, 8, 11, 13, 19} ← Referral

C = {2, 8, 11, 12, 18, 21} ← loyalty program

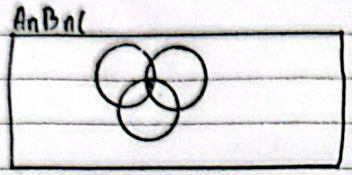
U = {1, 2, 3, ..., 21}

a. $A \cup B \cup C$



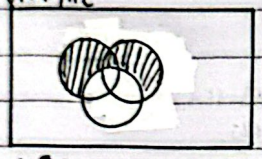
= {1, 2, 3, 4, 5, 6, 8, 11, 12, 13, 14, 16, 18, 19, 21}

$A \cap B \cap C$



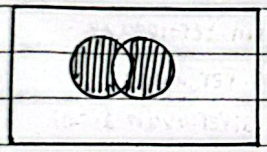
= {8, 11}

$(A \cup B) \cap C^c$



= {1, 3, 4, 5, 6, 13, 14, 16, 19}

$A \oplus B$



= {1, 3, 4, 5, 6, 13, 14, 16, 19}

c. $f(x) = 2x + 1$, $g(x) = 0.5x^2$

$f \circ g(x) = 2(0.5x^2) + 1 = x^2 + 1$

$g \circ f(x) = 0.5(2x+1)^2 = 2x^2 + 2x + 0.5$

- Tidak komutatif karena jika : $x = b + 2 = 3 \rightarrow f \circ g(3) = 10$

Sedangkan $g \circ f(3) = 24.5$, $g \circ f \neq f \circ g$

d. $f^{-1}(x) = \frac{x-1}{2}$

Pendapatan : $(K.100 + a + b) = 201$

volume $x = \frac{201-1}{2} = 100$

e. • $f(x) = 2x + 1$. karena berbentuk linear dengan gradien $\neq 0$. dan akan memiliki hasil yang unik pada setiap bilangan riil (injektif dan surjektif)

• $g(x) = 0.5x^2$. tidak injektif maupun surjektif. grafik ini memetakan bilangan negatif dan positif yang sama ke satu hasil yang identik.

SiDU

Dipindai dengan CamScanner